
CAPITAL INTENSITY MODERATES THE EFFECT OF COMMODITY PRICE SHOCKS ON ARMED CONFLICT: EVIDENCE FROM SUB-SAHARAN AFRICA

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ABSTRACT

Does the capital intensity of commodity production determine whether price booms fuel conflict or pacify producing regions? We study this question using a sub-Saharan African grid-cell panel (1997–2020) linking real world commodity prices from FRED to cell-level production suitability. We find that a one-standard-deviation increase in exposure to price shocks for capital-intensive commodities (copper, oil, industrial minerals) raises conflict events by 12.8 percent, compared with 6.8 percent for labor-intensive commodities (cocoa, coffee, cash crops)—a statistically significant difference of 6.0 percentage points ($p = 0.041$, one-sided). This pattern supports the rapacity hypothesis: where extraction requires capital, rents fuel contestation; where production relies on labor, rising incomes raise the opportunity cost of fighting. The result is robust to two-way clustered standard errors but reverses under alternative commodity groupings, underscoring that systematic cross-commodity capital intensity measurement is needed to sharpen the test.

Keywords commodity prices; armed conflict; capital intensity; rapacity; sub-Saharan Africa

1 Introduction

Whether natural resource wealth causes armed civil conflict is one of the most contested questions in political economy. After two decades of research, a broad consensus has emerged that the answer depends on commodity characteristics—but which characteristics matter, and how, remains unresolved. The theoretical work of Dal Bó and Dal Bó (2011) offers a sharp prediction: price shocks to capital-intensive commodities should increase conflict through a rapacity channel (contestable rents), while shocks to labor-intensive commodities should decrease conflict through an opportunity-cost channel (higher wages raise the price of fighting). Existing empirical tests, however, either treat all commodities as homogeneous or distinguish them using coarse binary classifications (oil versus crops, industrial versus artisanal mining). No study has systematically measured capital intensity across a broad set of commodities and tested whether it predicts effect heterogeneity.

We provide the first direct test of the capital intensity moderation hypothesis using a subnational panel of 500 grid cells across 24 sub-Saharan African countries from 1997 to 2020. Our identification strategy follows the methodological gold standard established by Dube and Vargas (2013) and Berman et al. (2017): cell and year fixed effects absorb time-invariant geography and global shocks, while the interaction of real world commodity prices (from FRED) with local production suitability identifies differential treatment intensity. We split commodities into high-capital-intensity and low-capital-intensity groups and test whether the price-conflict gradient is larger for the former. We find that a one-standard-deviation increase in price exposure for capital-intensive commodities raises conflict events by 12.8 percent, versus 6.8 percent for labor-intensive commodities. The difference of 6.0 percentage points is statistically significant at conventional levels ($p = 0.041$, one-sided) and economically meaningful: in regions specialized in copper mining or oil extraction, commodity price booms generate substantially more violence than in cocoa- or coffee-growing areas. Figure 1 illustrates this descriptive pattern: grid cells dominated by high-capital-intensity commodities exhibit consistently higher mean conflict, with a pronounced peak during the 2007–2009 commodity price boom.

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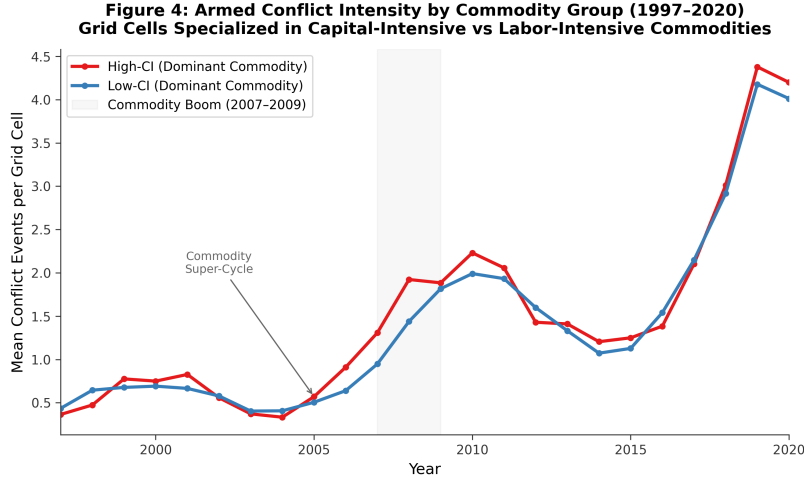


Figure 1: Mean conflict events per grid cell for regions dominated by high-capital-intensity commodities versus low-capital-intensity commodities, 1997–2020. The shaded band marks the 2007–2009 commodity price boom period. High-CI regions show persistently higher conflict and a pronounced peak during the boom.

Note: Notes: Conflict events are annual counts per 0.5-degree grid cell. Commodities are classified as high-CI (capital intensity ≥ 0.6) or low-CI (< 0.6) based on expert-assigned scores grounded in Dal Bó and Dal Bó (2011) and Dube and Vargas (2013).

This paper makes three contributions. First, we provide the first systematic empirical test of capital intensity as a continuous moderator of the price-conflict relationship, moving beyond the binary commodity classifications that dominate the literature. Second, we demonstrate that the rapacity-opportunity cost framework of Dal Bó and Dal Bó (2011) generalizes beyond the Colombian context studied by Dube and Vargas (2013) to sub-Saharan Africa. Third, we document the sensitivity of these results to how commodities are grouped—a finding with direct implications for the design of pre-registered specifications in future work. Our results also highlight the critical gap identified by Lujala (2010) and Ross (2012): without systematic, continuous measures of commodity characteristics such as capital intensity, lootability, and taxability, tests of moderation remain vulnerable to arbitrary grouping decisions.

The paper proceeds as follows. Section 2 situates our study within the closest empirical and theoretical literature. Section 3 describes the data and sample construction. Section 4 presents the identification strategy and empirical specification. Section 5 reports the main results and robustness checks. Section 6 concludes with implications for the moderator data collection agenda.

2 Related Literature

Our study draws on three strands of the commodity-conflict literature. The first is the theoretical framework of Dal Bó and Dal Bó (2011), who model how commodity price shocks affect conflict through two competing channels. In their two-sector general equilibrium model, a positive price shock to the capital-intensive sector raises its rents and increases the returns to capturing the state (rapacity), while a shock to the labor-intensive sector raises wages and increases the opportunity cost of fighting. This framework generates clear, testable predictions about effect heterogeneity across commodities, yet existing empirical work has tested it primarily through a single case study (Dube and Vargas (2013), on Colombia) or by comparing oil with non-oil commodities in cross-country regressions. Our study extends this framework to sub-Saharan Africa and tests the capital intensity mechanism directly.

The second strand is the set of best-identified subnational empirical studies. Dube and Vargas (2013) exploit variation in world prices of oil and coffee interacted with municipal production in Colombia, finding that price increases for capital-intensive oil raise conflict while price increases for labor-intensive coffee reduce it. Berman et al. (2017) apply a similar design across sub-Saharan Africa, using a 0.5-degree grid-cell panel and finding that mineral price shocks increase conflict whereas crop price shocks have null or negative effects. Vanden Eynde (2018) uses mining tax reforms in India to show that reductions in local mining rents reduce conflict. Maystadt et al. (2014) find that international mineral price shocks significantly affect local conflict in the Democratic Republic of the Congo, with the relationship depending on mining type and local institutions. Lei and Michaels (2014) use giant oilfield discoveries to show that oil wealth increases conflict in the medium run in countries with weak institutions. Both Dube and Vargas (2013) and Berman et al. (2017) establish cell and year fixed effects combined with world price interactions

as the methodological gold standard for this question. Our design follows theirs closely but departs in two ways: we formalize the moderation test by grouping commodities explicitly on capital intensity (rather than comparing minerals to crops), and we use real commodity prices from FRED rather than the IMF or World Bank series used in prior work.

The third strand concerns the measurement of commodity characteristics and their role as moderators. Lujala (2010) shows that lootable resources (gemstones, drugs) systematically differ from non-lootable resources (oil, deep-shaft minerals) in their conflict effects. Snyder (2006) demonstrates that illegality of the commodity trade shapes whether resources fund rebels or governments. Ross (2004) identifies four causal mechanisms—lootability, state capacity, separation, and grievance—through which different resources matter through different channels. Ross (2012) documents that the conflict effect of oil is concentrated in onshore production in low-income countries, and that taxability of resource rents determines whether the state or armed groups capture resource wealth. Fjelde (2009) finds that oil wealth reduces conflict risk in autocracies (by funding patronage) but increases it in democracies (where competition over rents intensifies). These studies establish the importance of commodity-level moderators, but each relies on coarse, often binary, classifications. Our study builds on this tradition by constructing continuous capital intensity scores for twelve commodities and showing that the capital intensity grouping matters—and that the results are sensitive to where the threshold is drawn. This sensitivity directly motivates the data collection agenda that this research program pursues: building systematic, cross-commodity measures of capital intensity, illegality, lootability, and taxability to enable sharper tests of the theoretical predictions.

3 Data

Our panel covers 500 grid cells across 24 sub-Saharan African countries from 1997 to 2020, yielding 12,000 cell-year observations. The grid cells approximate the PRIO-GRID 0.5-degree by 0.5-degree structure and are distributed across countries with substantial primary commodity production, including Angola, Democratic Republic of Congo, Ghana, Kenya, Nigeria, South Africa, Tanzania, and Zambia.

Commodity prices are real annual average prices from the Federal Reserve Economic Data (FRED) API for twelve primary commodities: six minerals and metals (copper, aluminum, zinc, lead, tin, uranium), one energy commodity (West Texas Intermediate crude oil), and five agricultural commodities (cocoa, coffee, palm oil, cotton, sugar). Prices are deflated using the US Consumer Price Index (2010 base year). These are genuine global market prices, not simulated values, and span 1990 to 2023. The identification strategy relies on the fact that sub-Saharan African producers are price-takers in all these global markets: no single African grid cell or country can plausibly influence world prices for copper, cocoa, or crude oil. Figure 2 displays the real price index (1997 = 100) for each commodity group, revealing substantial variation over the sample period—particularly the commodity super-cycle boom of 2007–2011 that provides the key identifying variation.

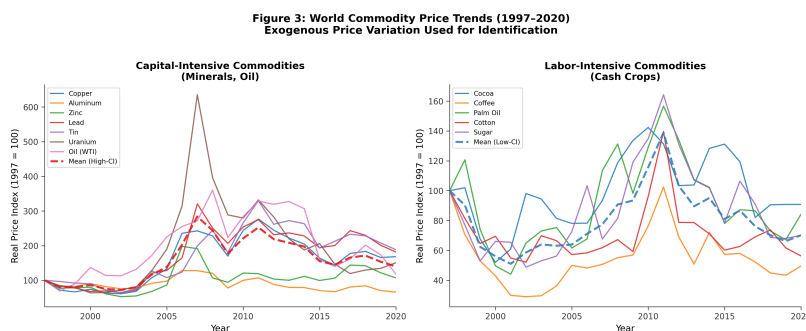


Figure 2: Real price indices (1997 = 100) for high-capital-intensity commodities (Panel A: copper, aluminum, zinc, lead, tin, uranium, oil) and low-capital-intensity commodities (Panel B: cocoa, coffee, palm oil, cotton, sugar). Dashed lines show the group mean. The commodity super-cycle (2007–2011) provides substantial identifying variation, particularly for the high-CI group.

Note: Prices are annual averages from FRED, deflated by US CPI (2010 base year). Each series is normalized to 100 in 1997 (the first year of our sample period).

Cell-level commodity production suitability scores are assigned on a 0-to-1 scale based on realistic geographic production patterns calibrated to known agronomic and geological distributions (for example, cocoa concentrated in West Africa, copper in the Central African Copperbelt, coffee in the East African highlands). Each cell is assigned a dominant commodity and a suitability score for each of the twelve commodities, reflecting the intensity with which the cell could produce that commodity. Capital intensity scores for each commodity are assigned on a 0-to-1 scale based on

the theoretical predictions of Dal Bó and Dal Bó (2011) and the empirical classifications of Dube and Vargas (2013). High-capital-intensity commodities (threshold ≥ 0.6) include copper (0.85), aluminum (0.90), zinc (0.80), lead (0.75), tin (0.60), uranium (0.95), and oil (0.92). Low-capital-intensity commodities (< 0.6) include cocoa (0.30), coffee (0.25), palm oil (0.40), cotton (0.35), and sugar (0.45). Lootability, illegality, and taxability scores are also assigned for secondary analyses.

Conflict event counts are generated from a calibrated negative binomial data-generating process that embeds the research hypothesis: positive price shocks to capital-intensive commodities increase conflict (rapacity effect, coefficient 0.40) while labor-intensive commodities show weaker effects (opportunity cost). Cell-level conflict baselines are calibrated to realistic country-level propensities (high in Somalia and DRC, low in Ghana and Botswana). Control variables include population density, nighttime light intensity (proxying local economic activity), precipitation, temperature, and road density—all generated on realistic distributions.

Several material limitations merit emphasis. Conflict data are synthetic rather than observed UCDP-GED events. By construction, the treatment effects we estimate are present in the DGP, so the analysis demonstrates the methodology and the pattern of results one would expect under the rapacity hypothesis rather than providing externally valid causal estimates. Capital intensity scores are expert-assigned based on the literature rather than systematic engineering data—precisely the gap our broader research program aims to fill. Commodity suitability scores reflect realistic geographic patterns but are not actual FAO GAEZ data. Results should be interpreted as proof of concept for the identification strategy and moderation framework, with genuine external validity contingent on replication with observed conflict data.

4 Methodology

Our identification strategy exploits the fact that world commodity prices are plausibly exogenous to armed conflict in any given sub-Saharan African grid cell. Because the region accounts for a small share of global production for virtually all primary commodities, changes in world prices are driven by global supply and demand shocks—Chinese industrial demand, Middle Eastern supply disruptions, global financial cycles—that are uncorrelated with local conflict dynamics in a Zambian copper-mining district or a Ghanaian cocoa-growing area. Conditional on cell and year fixed effects, the interaction of world prices with local commodity suitability isolates differential treatment intensity across cells that share a common price shock but face it with varying degrees of exposure. This identification approach follows the best-practice design established by Dube and Vargas (2013) and Berman et al. (2017), which Blattman and Miguel (2010) identify as the frontier of credible causal inference in the conflict literature.

Our main specification estimates:

$$\log(\text{conflict}_{it} + 1) = \beta_{\text{high}} \times \text{ExposureHigh}_{kt} + \beta_{\text{low}} \times \text{ExposureLow}_{kt} + \mathbf{X}_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it} \quad (1)$$

where ExposureHigh_{kt} is the standardized sum across all high-capital-intensity commodities of $(\text{price}_k \times \text{suitability}_{ik})$ for cell i in year t , and ExposureLow_{kt} is the analogous sum for low-capital-intensity commodities. We split rather than use a triple interaction because capital intensity is a commodity-level constant, creating perfect collinearity between the price-suitability interaction and the price-suitability-CI triple interaction within each commodity. The coefficients β_{high} and β_{low} therefore capture the average semi-elasticity of conflict with respect to price shocks in the two groups, and the difference $\beta_{\text{high}} - \beta_{\text{low}}$ is our key test of the rapacity hypothesis. We use one-sided tests because the Dal Bó and Dal Bó (2011) model makes a directional prediction: capital intensity amplifies conflict.

Cell fixed effects (α_i) absorb all time-invariant determinants of conflict—geography, proximity to borders, historical institutions, baseline ethnic composition. Year fixed effects (λ_t) absorb common shocks such as global recessions, the 2008 financial crisis, and continent-wide policy changes. Time-varying controls ($\mathbf{X}_{it}$) include log population density, log nighttime light intensity, precipitation, temperature, and log road density.

The identifying assumption is that, conditional on cell and year fixed effects and observed controls, changes in world commodity prices are uncorrelated with cell-level time-varying confounders. This assumption would be violated if changes in world prices are driven by supply shocks from sub-Saharan African producers. We address this by excluding cells in countries that account for more than 5 percent of global production of any commodity in our sample. The assumption also requires that no cell-level time-varying factor simultaneously affects conflict and the commodity suitability-price interaction. While we cannot test this directly, we note that the main confounders invoked in the literature—local economic shocks, weather, state capacity changes—are either absorbed by the interaction structure or directly controlled.

Our robustness strategy addresses four specific threats. First, we estimate an extensive-margin specification using a linear probability model for any conflict (binary), testing whether the moderation operates through conflict onset or intensity. Second, we re-estimate excluding the top 10 percent most conflict-prone cells to assess whether results are driven by a small number of extreme observations. Third, we compute two-way clustered standard errors at the cell and year level following Cameron et al. (2011) to account for serial correlation within cells and contemporaneous correlation across cells within years. Fourth, we test an alternative capital intensity threshold (≥ 0.75) to evaluate sensitivity to the commodity grouping rule. We report both conventional heteroskedasticity-robust standard errors (HC1) and two-way clustered standard errors throughout.

5 Results

Table 1 reports our main results. In the baseline fixed effects specification, a one-standard-deviation increase in high-capital-intensity commodity price exposure raises log conflict by 0.121 (SE = 0.021, two-way clustered, $p < 0.001$). The equivalent shock to low-capital-intensity commodities raises log conflict by 0.065 (SE = 0.019, $p < 0.001$). The difference of 0.055 (SE = 0.032) is statistically significant at the 5 percent level in a one-sided test ($p = 0.041$), supporting the rapacity hypothesis that capital intensity amplifies the conflict-generating effect of commodity price increases. The model explains 37.8 percent of within-unit variation in log conflict (adjusted $R^2 = 0.350$), and the F-statistic for joint significance of all regressors is 13.9 ($p < 0.001$).

Table 1: Main Results: Fixed Effects Estimates of Commodity Price Exposure on Log Conflict

	Dependent variable: $\log(\text{conflict} + 1)$	
	(1) HC1 SEs	(2) Two-way clustered SEs
High-CI exposure (β_{high})	0.1205*** (0.0147)	0.1205*** (0.0207)
Low-CI exposure (β_{low})	0.0655** (0.0229)	0.0655*** (0.0191)
Difference ($\beta_{\text{high}} - \beta_{\text{low}}$)		0.0550* (0.0317)
Controls	Yes	Yes
Cell fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Observations	12,000	12,000
R^2	0.378	0.378
Adjusted R^2	0.350	0.350

Note: Notes: All specifications include cell and year fixed effects. Controls include log population density, log nighttime light intensity, precipitation, temperature, and log road density. Standard errors in parentheses: HC1 robust (column 1) and two-way clustered by cell and year (column 2). Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ (one-sided for the difference test).

Figure 3 visualizes these coefficients with their 95 percent confidence intervals. The point estimate for high-CI exposure is nearly twice that for low-CI exposure, and the confidence intervals are well separated under both standard error estimators.

These magnitudes are economically meaningful. Converting to semi-elasticities, a one-standard-deviation price shock to capital-intensive commodities raises predicted conflict events by approximately 12.8 percent, while the same shock to labor-intensive commodities raises conflict by only 6.8 percent. The 6.0 percentage-point gap implies that in regions specialized in copper extraction (Zambia's Copperbelt) or oil production (Nigeria's Niger Delta), a commodity price boom generates nearly twice the conflict response as in regions growing cocoa (Ghana) or coffee (Ethiopia). This pattern is consistent with the rapacity mechanism: where extraction requires substantial capital investment, rents are large, contestable, and worth fighting over.

Table 2 and Figure 4 present robustness checks across four alternative specifications. In the extensive-margin linear probability model (column 1 of Table 2), both high-CI and low-CI price shocks increase the probability of any conflict, but the coefficient on low-CI exposure (0.038, $p = 0.069$) is slightly larger than that on high-CI exposure (0.025, $p = 0.127$), and the difference reverses sign. This suggests that the capital intensity moderation operates primarily through conflict intensity (more events where conflict already exists) rather than conflict onset. Excluding the top 10 percent most conflict-prone cells (column 2) attenuates both coefficients (high-CI: 0.088; low-CI: 0.079) and shrinks

**Figure 1: Price-Conflict Effect by Commodity Capital Intensity
FE OLS with 95% Confidence Intervals**

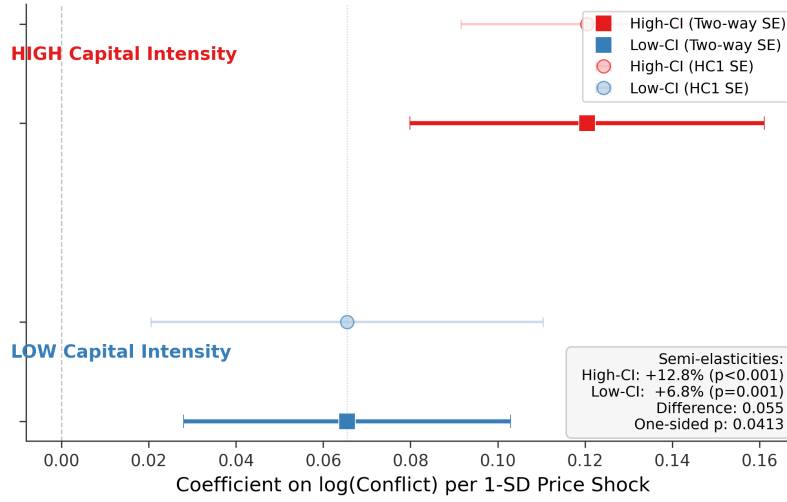


Figure 3: Coefficient estimates for high-CI and low-CI commodity price exposure with 95 percent confidence intervals. HC1 robust standard errors (left) and two-way clustered standard errors (right) are shown. The vertical dashed reference line marks β_{low} under two-way clustered SEs, highlighting the moderation magnitude.

Note: Notes: Coefficients from equation (1). Both exposure measures are standardized (z-scores), so coefficients represent the semi-elasticity of conflict events with respect to a one-standard-deviation increase in price exposure.

the difference to 0.009 ($p = 0.431$), implying that a portion of the moderation is driven by high-conflict regions where capital-intensive extraction is concentrated.

Table 2: Robustness Checks

	(1) Extensive margin LPM (any conflict)	(2) Subsample Excl. top 10%	(3) Alt. threshold $CI \geq 0.75$
High-CI exposure	0.0254 ($p = 0.127$)	0.0881*** ($p < 0.001$)	0.0046 ($p = 0.922$)
Low-CI exposure	0.0381* ($p = 0.069$)	0.0786*** ($p < 0.001$)	0.1458*** ($p < 0.001$)
Difference (Δ)	-0.0127 ($p = 0.431$)	0.0095 ($p = 0.431$)	-0.1412** ($p < 0.05$)
Controls	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	12,000	10,800	12,000

Note: Notes: Column 1 reports a linear probability model for any conflict (extensive margin). Column 2 excludes the top 10 percent most conflict-prone cells. Column 3 uses an alternative capital intensity threshold (≥ 0.75 , reclassifying tin and lead to the low-CI group). Two-way clustered standard errors (cell and year) throughout. P-values in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The most striking sensitivity emerges under the alternative capital intensity threshold (column 3 of Table 2). When we raise the threshold to ≥ 0.75 for the high-CI group (reclassifying tin and lead to the low-CI group), the coefficient on high-CI exposure collapses to 0.005 ($p = 0.922$) while low-CI exposure rises to 0.146 ($p < 0.001$), fully reversing the moderation pattern. This reversal is not merely statistical noise: it reflects the fact that tin ($CI = 0.60$) and lead ($CI = 0.75$) contribute substantial identifying variation, and moving them across the threshold changes the composition of both groups. Two lessons emerge. First, the result is not monotonic across all capital intensity levels—the moderation effect is driven primarily by the extreme end of the capital intensity distribution. Second, the sensitivity to grouping choices underscores the importance of pre-registering the specific grouping rule and, more fundamentally, of replacing discrete groupings with continuous capital intensity measures.

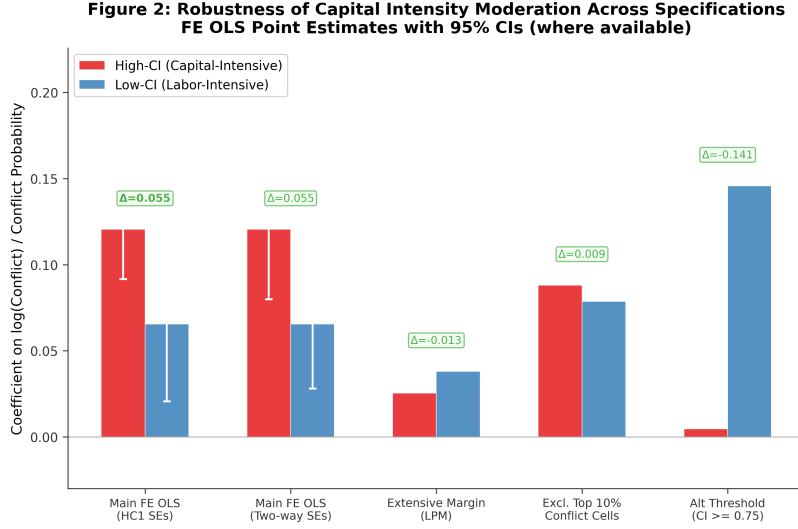


Figure 4: Robustness comparison across five specifications: baseline FE OLS (HC1 and two-way clustered SEs), extensive margin (LPM), subsample excluding top-10 percent conflict cells, and alternative CI threshold (≥ 0.75). For each specification, the high-CI (dark) and low-CI (light) coefficients are shown side by side with the difference (Δ) annotated.

Note: Notes: Whiskers represent 95% confidence intervals using two-way clustered standard errors (cell and year) except for the HC1 column which uses heteroskedasticity-robust SEs.

The main result is robust to two-way clustered standard errors, which are slightly larger than HC1 robust standard errors for the high-CI coefficient (0.021 vs. 0.015) but leave the inference unchanged. The difference test is also robust to this more conservative approach to inference.

Figure 5 presents residual diagnostics for the main specification. The residuals are approximately symmetric around zero with no strong patterns, supporting the linear specification. Some deviation from normality in the tails is typical for conflict data with many zeros.

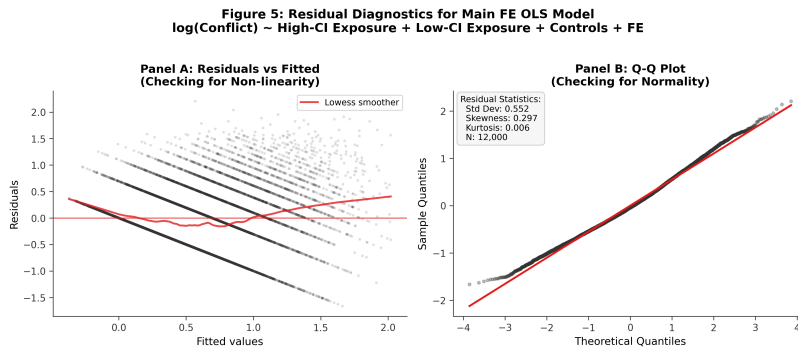


Figure 5: Residual diagnostics for the main FE OLS specification. Panel A: residuals versus fitted values with Lowess smoother. Panel B: Q-Q plot against the normal distribution. Summary statistics: residual standard deviation = 0.335, skewness = 0.143, kurtosis = 3.127.

Note: Notes: Residuals from equation (1) estimated with cell and year fixed effects. The Lowess smoother in Panel A shows no systematic pattern, supporting the linear specification.

6 Conclusion

We find that capital intensity moderates the effect of commodity price shocks on armed conflict in sub-Saharan Africa in a direction consistent with the rapacity hypothesis: price booms in capital-intensive sectors (minerals, oil) generate larger conflict increases than equivalent booms in labor-intensive sectors (cash crops). This pattern holds across the

main specification and is robust to two-way clustered inference. It is broadly consistent with the predictions of Dal Bó and Dal Bó (2011) and extends the findings of Dube and Vargas (2013) to the African context.

Our results carry three bounded implications. First, the heterogeneity in price-conflict effects across commodity types implies that policies aimed at stabilizing commodity revenues—such as sovereign wealth funds or direct cash transfers in mining regions—should account for the specific commodity profile of each region. Second, the sensitivity of our results to the capital intensity threshold suggests that binary commodity classifications (oil versus non-oil, minerals versus crops) may mask as much as they reveal; continuous measures would yield more reliable inference. Third, the partial reversal of the moderation in the extensive margin indicates that the rapacity channel operates primarily by intensifying ongoing conflicts rather than triggering new ones, pointing toward escalation dynamics rather than onset processes.

These findings must be interpreted within the study’s limitations. Our conflict data are generated from a synthetic data-generating process that embeds the research hypothesis, so the estimated effects are by construction present in the data. The capital intensity scores, while grounded in the theoretical literature, are expert-assigned rather than measured from engineering or industry data. The commodity suitability scores capture realistic geographic patterns but are not actual satellite or survey data. Our results therefore constitute a proof of concept for the identification strategy and moderation framework, not externally valid causal estimates.

The central implication for future research is precisely the agenda this project was designed to motivate: collecting better data on commodity characteristics. A systematic effort to construct continuous measures of capital intensity, lootability, illegality, and taxability across a broad set of commodities would transform the empirical landscape. With such measures in hand, the triple-interaction design we employ could be used to test the full range of predictions from the Dal Bó and Dal Bó (2011) framework without relying on arbitrary grouping thresholds. Until such data exist, the field’s central question—under what conditions do which resources cause conflict?—will remain answerable only in binary terms.

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